

Testing Plan

Sprint 2

FIT2101 Team 5

Introduction:

The testing plan being conducted will be focused on the tasks allocated to group members in sprint two. This testing plan will document each task's requirements and input/output expected of each task. This testing plan will also clearly outline how the test can be easily reproduced by the user.

User stories:

This lists all user stories that will be incorporated into the tasks created, giving reasoning for tasks that were created and enacted upon:

THEME: Task Popup:

- As a user, I want to be able to add a team member from a drop-down list to a particular task so that I can assign tasks to every team member and keep track of it.
- As a user, I want the priority of each task to be colour-coded so that I can differentiate between tasks visually
- As a user, I want to be able to differentiate tasks by the colour of the tag so I know what 'skills' are required for the task

THEME: Sprints

- As a user, I want to be able to view details of a sprint once I click on it, so that I can keep track of all requirements in a particular sprint.
- As a user, I want to be able to move tasks from a Sprint to the Product Backlog, so that I can implement the said task in an upcoming sprint.
- As a user, I want to be able to create and delete sprints so that I am aware of all sprints occurring during the project timeframe.

THEME: Team Management

- As a user I want to be able to have a dashboard with a date range to see how my team is performing during a particular time frame to keep track of how each team member is progressing.
- As a user, I want to be able to click into each individual team member in the team dashboard to check their contribution in the form of a graph, showing how many hours they have worked in a specified date range, so that I know that they are progressing through the project at a steady pace.

Testing process:

The testing processes that will be used to test new implementation on the website will be primary based on functional testing and UI testing. This is due to the tasks of the sprints being based on new functionality implementation and the ease of navigation for the user to perform tasks. Tasks are labelled with the testing process they will be conducted with in the task testing analysis portion below.

Task testing analysis:

THEME: Task Popup:

Assign team member:

When assigning a team member to a task, we want to be able to select a team member from a list of team members through a drop-down list, this allows easy team management through simple navigation to the assignment of new members.

UI testing: The drop-down menu is easily recognizable and interactable for the user, once interacted with, the user will see a list of team members available to be assigned to the task.

Functionality testing: There should be a dropdown menu tied to a task. Once the user interacts with the drop-down menu, the user should be presented with a list of initialised team members

Recreate test:

1. Open the product backlog page
2. Create a new task from the top right 'plus' button
3. Click the 'Assignee' dropdown menu
4. Select a team member to assign to the task
5. Save new task
6. Click the edit task button from the product backlog page, represented by a pen icon
7. See if the assignee field is the same as previously selected

Priority colours:

When tasks are created with a priority label, the different thresholds of priority should be easily distinguishable not only by name but by colour. This allows better visual cues and usability when interacting with the priority label.

UI testing: When a task is created with a priority label, the priority label will be highlighted with grey for 'Low', yellow for 'Medium', orange for 'High' and red for 'Critical'.

Recreate test:

8. Open the product backlog page
9. Create a new task from the top right 'plus' button

10. Choose a priority for the new task
11. See if the priority label is correctly assigned to the colour code

Tag colours:

When tasks are created with a tag label, the different areas of expertise should be easily differentiable not only by name but by colours. This allows better distinguishability on which areas of expertise and skill needed are for each task.

UI testing: When a task is created with a tag label, the tag label will be highlighted with green for 'Testing', light purple for 'User Interface' and red for 'Core'.

Recreate test:

1. Open the product backlog page
2. Create a new task from the top right 'plus' button
3. Choose a tag for the new task
4. See if the priority label is correctly assigned to the colour code

THEME: Sprints:

View details of sprint:

When an instance of a sprint has been created, the sprint's details should be easily accessible by clicking on the sprint itself, presenting the user with its contents.

Functionality testing: The left side of the home will list all created instances of a sprint, when clicking an individual sprint from the list on the left of the screen, it will navigate the user to the sprint page with all details of the sprint

Recreate test:

1. Click on a pre-existing sprint on the left side of the page (if no sprints are present, create a new sprint by following 'Creating and deleting sprints' task's instructions)
2. See if the correct sprint has been navigated to and the details are correctly presented on the page

Moving tasks from product to sprint backlog:

When tasks are created in the product backlog, they should be able to be moved to the sprint backlog by users to dedicate tasks to a sprint.

UI testing: When a task had been moved from the product backlog, it should be visible on the designated sprint page for the user to see.

Functionality testing: From the product backlog page, a button on the row of a specific task will allow the user to move the task into a pre-existing sprint

Recreate test:

1. Create a task from on the product backlog page by clicking the plus button on the top right
2. Click the move to sprint icon labelled as a folder with an arrow icon
3. Choose a pre-existing sprint to move the task to
4. Navigate to the sprint by clicking on the sprint located on the left of the screen
5. See if the task has been moved to the specific sprint

Creating sprints:

The user should be able to create and allocate sprints that can be later accessed and edited to add additional sprint details, allowing easy updates to sprints.

UI testing: When a new sprint has been created, it should be visible in the list of sprint pages.

Functionality testing: There should be a button on the left side of the page that allows users to click to add a new instance of a sprint. When a new sprint has been created, it should be accessible by clicking on the individual sprint and bringing the user to the new sprint page.

Recreate test:

1. Click the new sprint button on the left side of the page, this should bring up a pop-up window
2. Enter a name for the sprint and the date from start to finish of the sprint into the pop-up window's input options
3. Click the 'ADD' button after details have been filled
4. Click the sprint on the left side of the screen (The sprint will be denoted by its name)
5. See if the page is navigated to the correct sprint with the correct corresponding details

THEME: Team management:

Time log dashboard:

A time-log dashboard should be present to allow users easy tracking of each team member's progression during each sprint and their time spent working on each task/sprint workload.

UI testing: When the user has traversed to the time-log dashboard, the page should display a date range for a specified timeframe that shows documented time each team member has spent on their work.

Recreate test:

1. Click on the team button on the left of the page to traverse to the team time log dashboard
2. Change the start and end dates above the graph to look through specified time frames

Contribution details of a team member:

The time-log dashboard should offer traversal to individual team members' contribution information. The contribution information should offer a graph to the user that documents the hours they have worked in a specified date range.

Functionality testing: When selecting a specific team member in the time log dashboard, the time log graph should filter to the selected team member's statistics

Recreate test:

1. Click on the team button on the left of the page to traverse to the team time log dashboard
2. In the row of a team member, click the checkbox on the left to filter the graph to only show that team member's statistics
3. Click the checkbox in the first row to select all team members, this should show all team member's statistics on the graph

Program setup instructions:

With access to the git repository, the user can run and test the program by following these steps.

1. Open an IED
2. Clone git repository
3. Open terminal (current directory is the 'Projects' folder)
4. Type in terminal: npm install
5. Type in terminal: npm start